

ID:

Name:

Brac University

Semester: Spring 2026

Course Code: CSE250

Circuits and Electronics

Set

A

Assessment: *Final Exam*

Duration: 1 hour 40 minutes

Date: May 19, 2026

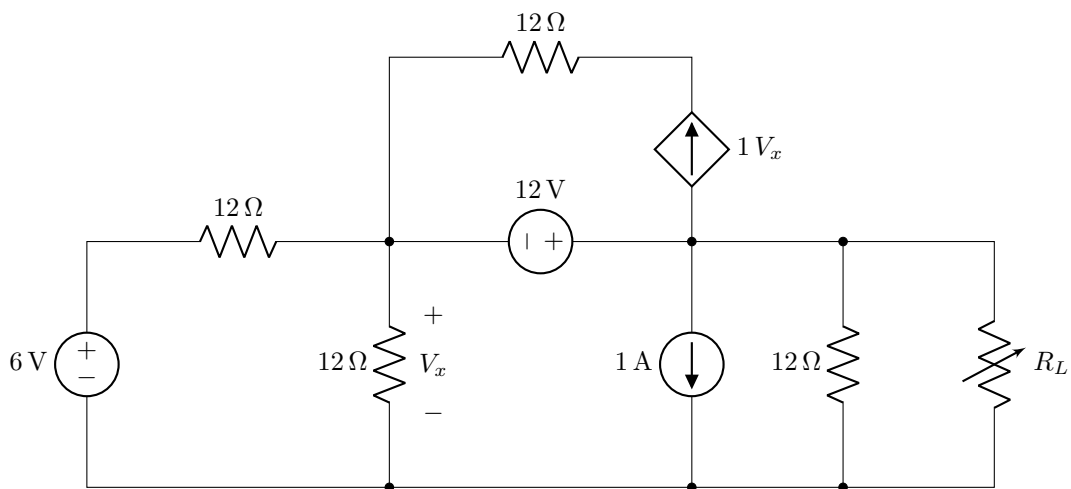
Full Marks: 55

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
- ✓ All **5 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
- ✓ Symbols have their usual meanings.

■ Question 1 of 5

[CO2] [16 marks]

For the circuit shown below,

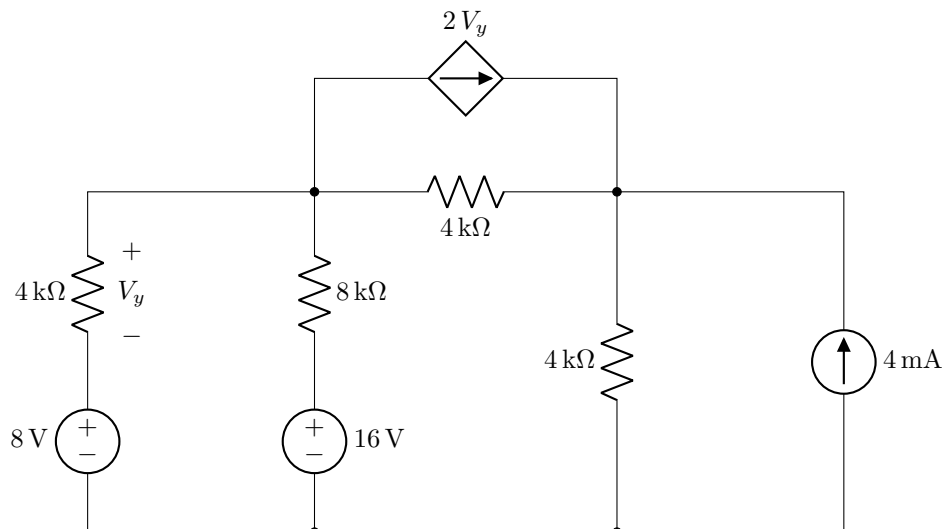


Apply **Thevenin's Theorem** or **Norton's Theorem** to determine the **maximum power** that the circuit can deliver to the load resistance R_L .

■ Question 2 of 5

[CO2] [12 marks]

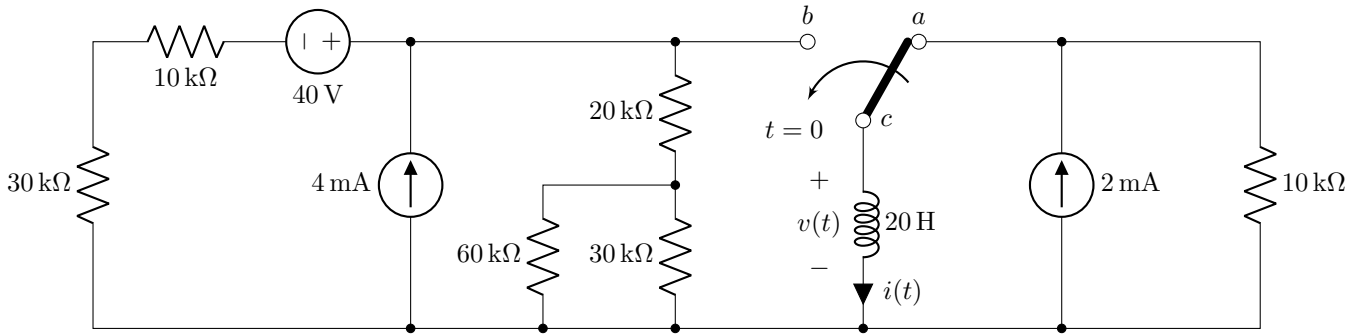
Apply successive **Source Transformations** to reduce the circuit shown below to a single loop and **determine V_y** .



■ Question 3 of 5

[CO3] [16 marks]

The switch in the circuit shown below is moved from position *a* to position *b* at $t = 0$, while position *c* remains fixed.



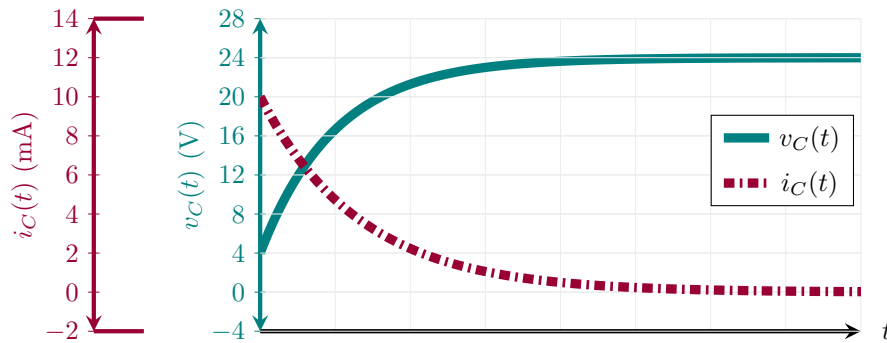
Analyze the **Transient Behavior** of the circuit to answer the following questions:

- [13 marks] Determine the inductor current $i(t)$ as a function of time for $t > 0$.
- [3 marks] Determine the inductor voltage $v(t)$ as a function of time for $t > 0$.

■ Question 4 of 5

[CO3] [5 marks]

The voltage and current responses of a $1\ \mu\text{F}$ capacitor in an RC circuit are plotted below as a function of time t , where $v_C(t)$ and $i_C(t)$ denote the voltage across and current through the capacitor, respectively. Note that v_C and i_C are plotted on separate y-axes.

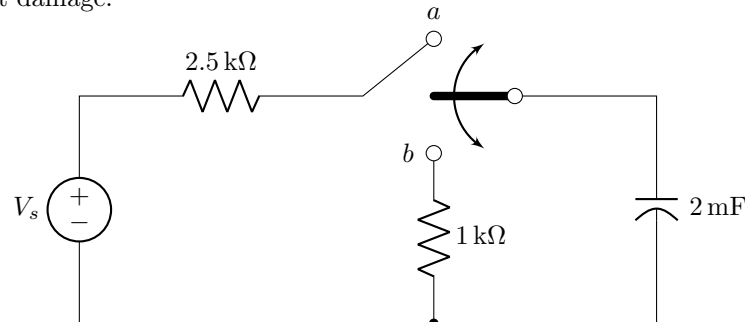


- [1 mark] Write an equation for the voltage v_C across the capacitor in terms of time t and the time constant τ .
- [2 marks] Using the equation for v_C from part (a), write an equation for the current i_C through the capacitor in terms of time t and the time constant τ .
- [2 marks] **Determine** the time constant τ of the circuit.

■ Question 5 of 5

[CO3] [6 marks]

When the switch in the following circuit is at position *a*, the capacitor charges. When moved to position *b*, it discharges. The capacitor stores a maximum energy of 25 mJ when fully charged at position *a*. However, the capacitor can safely retain a maximum of 16 mJ without damage.



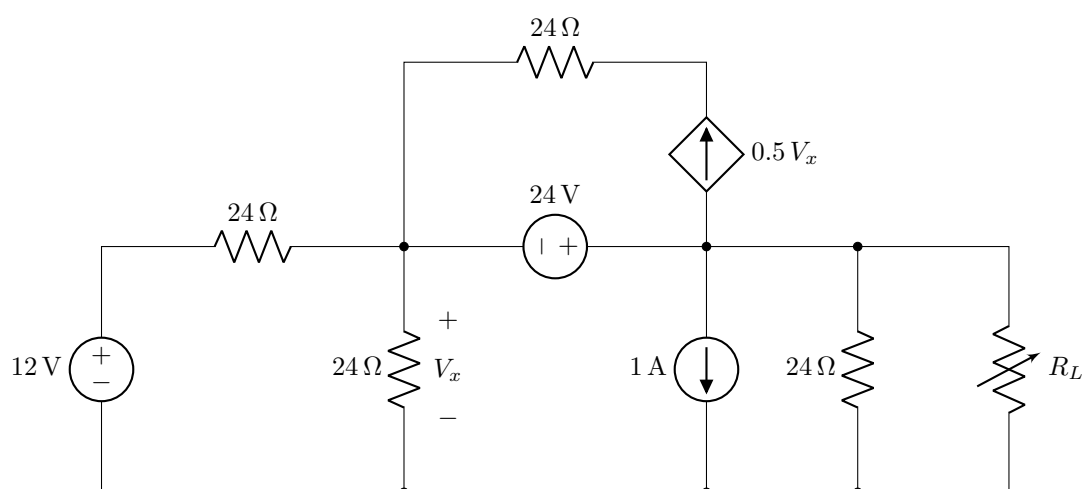
- [1 mark] **Determine** the source voltage V_s .
- [1 mark] What is the maximum voltage that can be applied across the capacitor without exceeding the safe energy limit, that is, 16 mJ?
- [3 marks] **Determine** the maximum time the capacitor can remain at position *a* before its stored energy exceeds the safe limit. Assume the capacitor is initially uncharged when connected to position *a*.
- [1 mark] **Determine** the time required for the capacitor to fully discharge when switched to position *b*.

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- ✓ All **5 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
- ✓ Symbols have their usual meanings.

■ Question 1 of 5

[CO2] [16 marks]

For the circuit shown below,

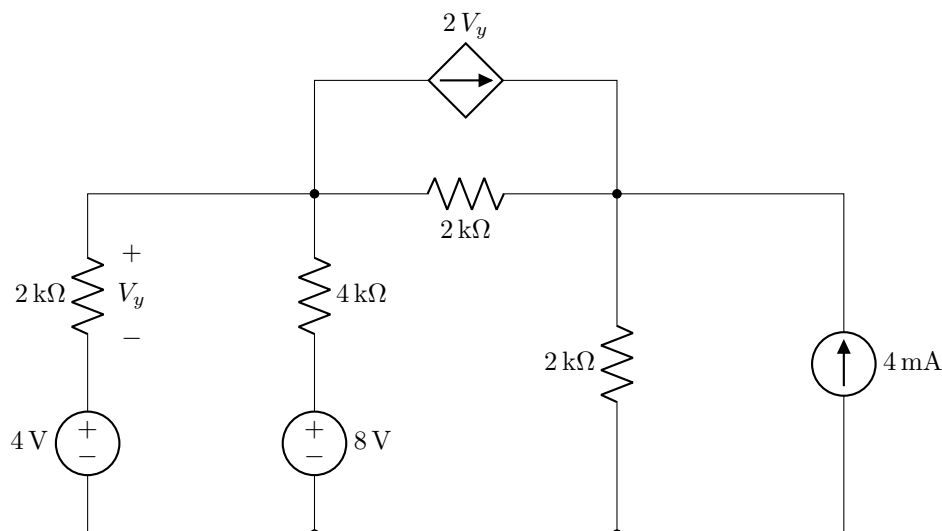


Apply **Thevenin's Theorem** or **Norton's Theorem** to determine the **maximum power** that the circuit can deliver to the load resistance R_L .

■ Question 2 of 5

[CO2] [12 marks]

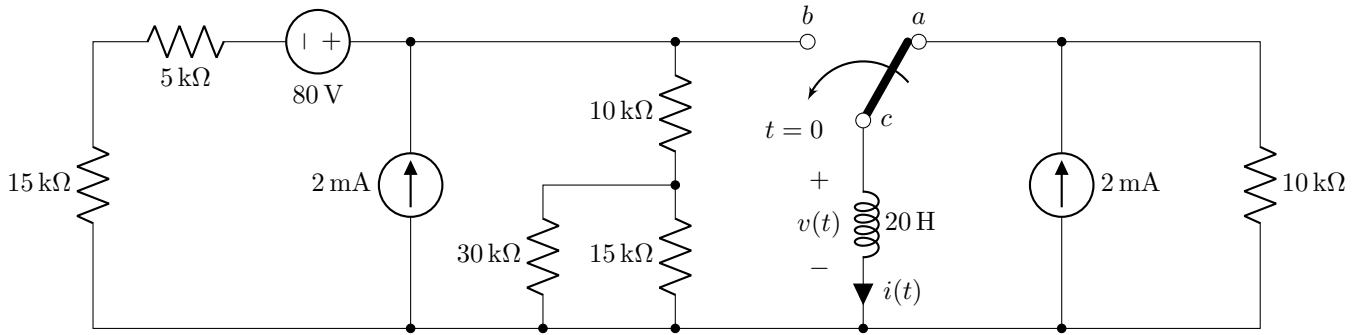
Apply successive **Source Transformations** to reduce the circuit shown below to a single loop and **determine** V_y .



■ Question 3 of 5

[CO3] [16 marks]

The switch in the circuit shown below is moved from position a to position b at $t = 0$, while position c remains fixed.



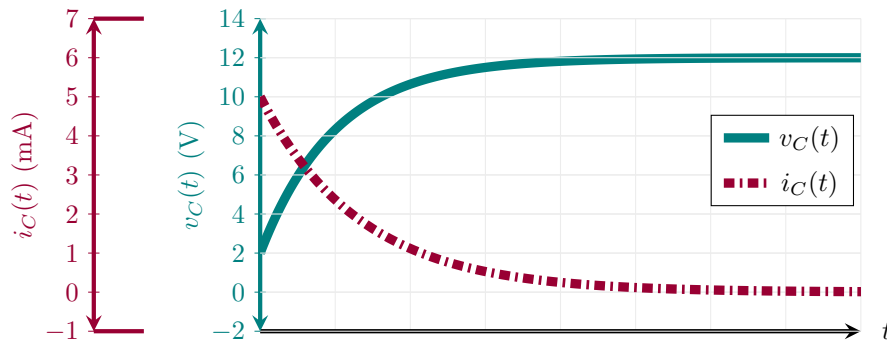
Analyze the **Transient Behavior** of the circuit to answer the following questions:

- [13 marks] Determine the inductor current $i(t)$ as a function of time for $t > 0$.
- [3 marks] Determine the inductor voltage $v(t)$ as a function of time for $t > 0$.

■ Question 4 of 5

[CO3] [5 marks]

The voltage and current responses of a $1 \mu\text{F}$ capacitor in an RC circuit are plotted below as a function of time t , where $v_C(t)$ and $i_C(t)$ denote the voltage across and current through the capacitor, respectively. Note that v_C and i_C are plotted on separate y-axes.

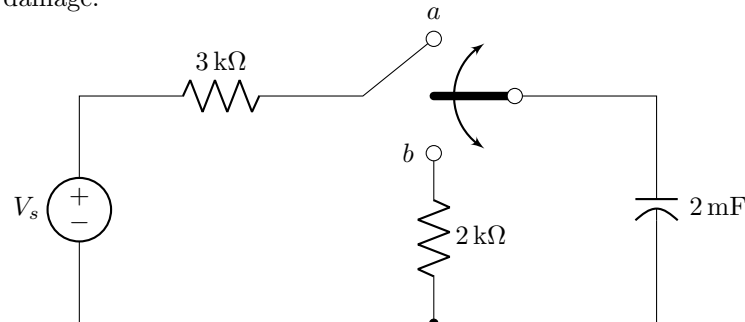


- [1 mark] Write an equation for the voltage v_C across the capacitor in terms of time t and the time constant τ .
- [2 marks] Using the equation for v_C from part (a), write an equation for the current i_C through the capacitor in terms of time t and the time constant τ .
- [2 marks] **Determine** the time constant τ of the circuit.

■ Question 5 of 5

[CO3] [6 marks]

When the switch in the following circuit is at position a , the capacitor charges. When moved to position b , it discharges. The capacitor stores a maximum energy of 25 mJ when fully charged at position a . However, the capacitor can safely retain a maximum of 9 mJ without damage.



- [1 mark] **Determine** the source voltage V_s .
- [1 mark] What is the maximum voltage that can be applied across the capacitor without exceeding the safe energy limit, that is, 9 mJ ?
- [3 marks] **Determine** the maximum time the capacitor can remain at position a before its stored energy exceeds the safe limit. Assume the capacitor is initially uncharged when connected to position a .
- [1 mark] **Determine** the time required for the capacitor to fully discharge when switched to position b .